

Arunalu: Learning Ecosystem to Overcome Sinhala Reading Weakness Due To Dyslexia

Lakshani Sandathara
Department of Software Engineering
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
dmlakchani@gmail.com

Harini Hapuarachchi
Department of Software Engineering
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
harini.upendrika@gmail.com

Shalini Tissera
Department of Information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
shalinitissera2@gmail.com

Samantha Thelijagoda
SLIIT Business School
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
samantha.t@slit.lk

Ransi Satharani
Department of Information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
ransiambagahage@gmail.com

Abstract— Dyslexia is an impairment in ability in reading. People having Dyslexia has difficulties in identifying specific letters and words and identifying speech sounds and decoding the letters which leads to difficulties in comprehension, spelling and writing. Dyslexia may severely affects language development and impacts reading and other language based improvement and functioning. “ARUNALU: Learning ecosystem to overcome reading disabilities in Sinhala language due to Dyslexia” has been proposed as a multi-sensory mobile solution, in native language of Sri Lanka (Sinhala) and with effective screening and intervention methodologies recommended by health professionals. Objective is to deliver, a phonological support to enhance reading skills of dyslexic children by providing a machine learning based automated screening and intervention mobile solution. Through these reading environments, there's a reward system in intervention process to encourage the user, and also users and respective parties can analyze user's progress. The proposed system is mainly based on Voice recognition, Natural Language Processing, Machine Learning and Deep Learning concepts collaborating with reading and gaming environments. Core Objective of the proposed system is to come up with a better and effective screening and intervention methodologies for early identification of Dyslexia and provide phonological training to overcome Sinhala reading difficulties due to Dyslexia in a user friendly manner

Keywords— *Dyslexia, Natural Language Processing, Voice recognition, mobile solution, Reading weakness*

I. INTRODUCTION

Dyslexia is not a disease [1]. It cannot be cured by drugs. [2] It causes due to reduced function of one area towards the back of the brain called, occipito-temporal cortex. Dyslexia may severely affect language development and impacts reading and other language-based improvement and functioning [3] According to the previous research work, [4] reveals that phonological training has the potential in making the cortical differences. [5] Phonological training that may apply for

children who are at risk of Dyslexia are letter-sound training, phonological awareness training, and both. [6] At the early stage through processing the distinctive sound elements that constitute a language, the remedy as phonological training can be done. There are three forms of developmental dyslexia namely,

(a) Phonological dyslexia, (b) Surface dyslexia and (c) Mixed dyslexia. Phonological dyslexia is having four categories, [7]

- I. Auditory hypothesis – poor performance on a number of auditory tasks (tone discrimination, temporal order judgment, repetition tasks, backward masking, and difficulties in perceptual discrimination of speech sounds.)
- II. Visual hypothesis – observation of visual deficits Ex: - visual stress, bio- ocular fixation instability, visual- attentional disorder
- III. Magnocellular hypothesis – integrates the preceding theories (auditory and visual) in defending a theory.
- IV. Cerebellar hypothesis – motor deficit, the poor capacity of motor coordination, and problems of equilibrium

40% of bad readers have an auditory deficit. A minority have visual and motor deficit. Therefore, phonological deficit constitutes a common factor in the development of dyslexia [7]. Although still not fully understood it is recognized that what most dyslexics have in common is a difficulty in grasping the shapes of letters and then relating those shapes to the sounds that the letters symbolize [8].

ARUNALU is a learning ecosystem to overcome Sinhala reading weakness due to dyslexia. This solution is mainly targeting on local children context in age 5 –6 approximately 1-33% of children in Sri Lanka [3] .In this solution, mainly focusing on improving the reading ability via four screening and four solution methodologies. Therefore, it will be a benefit for their educational and social engagements in future. ARUNALU is mainly focusing on, improve the reading skills which caused due to Dyslexia providing phonological training through, Vocal pattern analysis through letters and words, Feature extraction from Sinhala letters through reading and voice pattern analysis, Vocal pattern analysis through motion pictures and screening and intervention through voice pattern analysis via Story Building concept. This study focusing on filling the gap in existing screening and intervention applications for dyslexia.

II. BACKGROUND

Existing solutions in Sri Lanka, [9] is the first solution came up in native language with screening and intervention of Dyslexia and another to disabilities named Dysgraphia and Dyscalculia. The solution was a tablet computer game series. This solution is screen learning disabilities in age within 4 - 6 kids via tablet computer game series, kinetic based game series and website. “The Hope” is an Interactive Mobile Solution to Overcome the English Writing, Reading and Speaking Weaknesses of Dyslexia which assembles Dyslexia therapies in a graphical and creative way to present the user to make it more accurate. [10], but native language support was not there for the local context. Study in [11] reveals that dyslexic children can be treated by therapy sessions. Currently, the approach is a manual procedure and the solution was a digitalized mobile solutions for conducting the therapies. But the solution mainly focusing on English letters. “Dyseggia”, a game application with word exercises for children with dyslexia which received positive feedback [12]. EasyLexia: A Mobile Application for Children with Learning Difficulties by providing set of games [13]. According to findings [14] Gamification, game- thinking is an effective and efficient motivational tool that helps the children in overall e-learning process. Studies [15] reveal that use of gamification concepts can be applied to motivate dyslexic children by applying gamification as an educational tool and a motivational tool. [16] Recent approach provides screening and intervention for Dyslexia in native language, which provides one methodology. Table 1 contains a comparison between existing products and the proposed system.

After analyzing existing systems, there are limitations in the technology, age group and the screening and intervention methodologies used. And supporting of better automated screening methods are lacking. Furthermore analysis indicated that giving a special care on reading difficulties and specific solutions for reading difficulties are lacking. And ARUNALU has considered native language support, early identification, user friendliness and also providing four screening and intervention methods for unique identification

Feature	EasyLexia	Dyslexia Baca	YALU	The Hope	ALEXZA	ARUNALU
Dyslexia screening	×	×	✓	✓	×	✓
Machine Learning	×	×	×	✓	✓	✓
Gaming Environment	✓	✓	✓	✓	×	✓
Dyslexia intervention	✓	✓	×	✓	✓	✓
Support Sinhala language	×	×	✓	×	×	✓
Letters and word recognition	✓	✓	✓	✓	✓	✓
Voice recognition	✓	✓	✓	✓	✓	✓
Progress analysis through statistics	×	×	✓	✓	✓	✓
Mobile solution	✓	✓	×	✓	✓	✓
Focused on specially Reading ability enhancement	×	✓	×	×	×	✓
Monitoring by multiple users	×	×	×	✓	×	✓

Table 1: Comparison between existing solutions and ARUNALU

III. METHODOLOGY

In the solution, it provides four screening and four intervention methods which difference from the functionality. The reason for providing four screening and intervention methods for identifying the user is Dyslexic or not in a unique manner.

- I. Vocal Pattern analysis through letters and words
- II. Feature extraction from Sinhala letters through reading and voice pattern analysis
- III. Vocal pattern analysis through motion pictures
- IV. Screening and Intervention through voice pattern analysis via Story Building concept

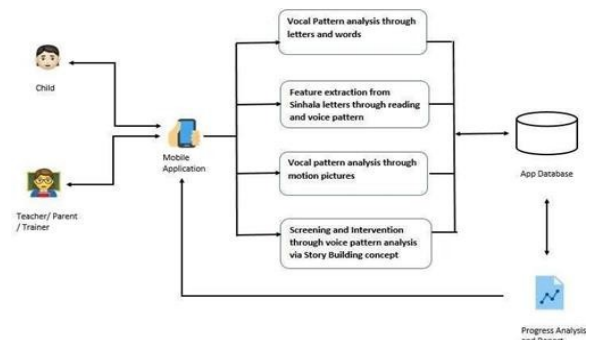


Figure 1: System Overview Diagram of ARUNALU

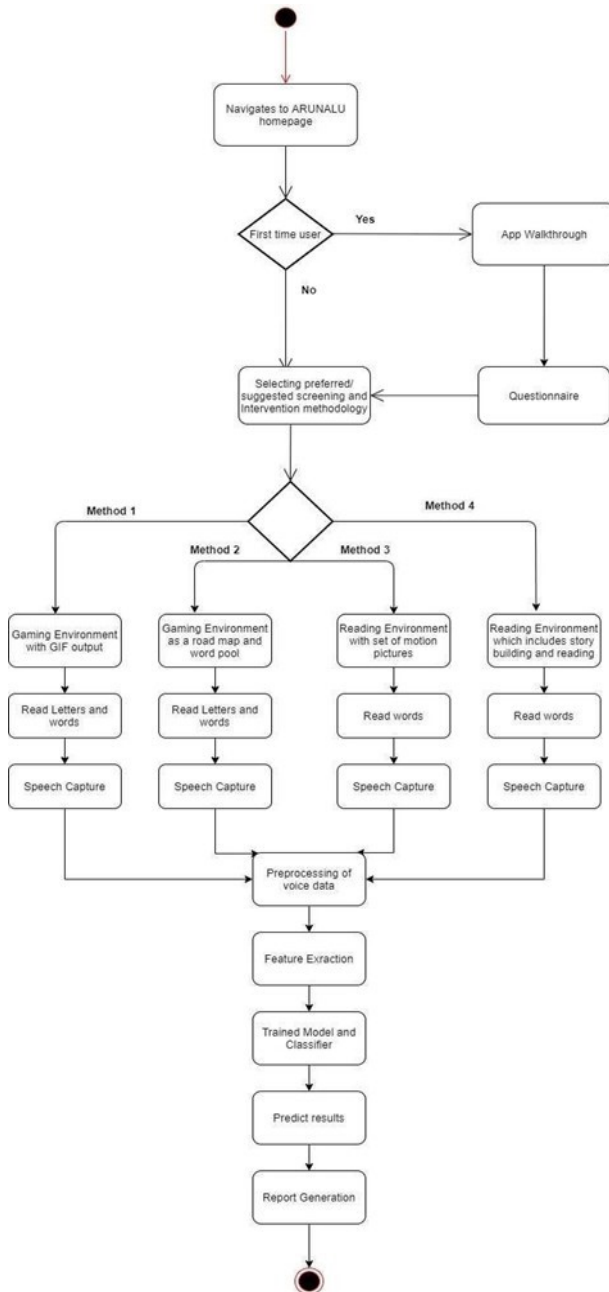


Figure 2: Activity flow of ARUNALU

Before selecting a screening methodology, first time user need to answer a questionnaire to get data whether the person is having other reading difficulties other than Dyslexia (Problem in Vision, Problems in responding to rhythmic reading approaches, etc.). Then the application recommending a method for user, or else user can select a preferred methodology provided in the application.

A. Vocal Pattern analysis through letters and words (Gaming Environment with GIF output)

In this component, screening and intervention is done using an interactive gaming environment. Reading test will run in stage by stage to overcome the weaknesses since the game load the user the stages dynamically.

Method and Solution for Dyslexia remain series of pronunciation tests. Simply define three categories namely First Step, Second Step and Third Step. Then for corresponding categories specific predefined words are well- defined. First Method, when there are two letters if the first letter is pronouncing correctly only it highlights and move to the next letter and after pronouncing hold word correctly the user can see a GIF related to the word.

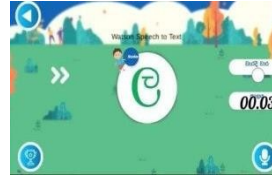


Figure 3: Gaming environment for screening process using GIF output method



Figure 4: Gaming environment for screening process using GIF output method

B. Feature extraction from Sinhala letters through reading and voice pattern analysis (Gaming Environment as a road map and word pool)

Among different type of screening methods and intervention methods, Gamification is the best way to improve the learning experience and the performance of students [17].

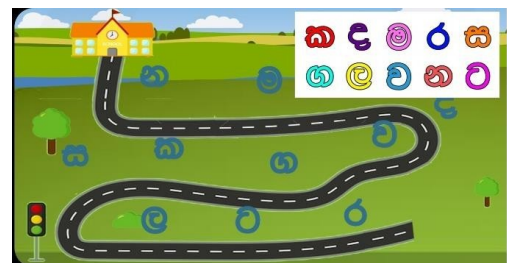


Figure 5: Dyslexia Screening and Intervention using road map and word pool

This component is using a gaming environment to deliver the screening and intervention process. The system displays a road map with letters and letter pool (Figure 5). It is a multi-sensory environment which allows user to identify Sinhala letters and pronunciation. User should correctly identify the letters and pair them by drag and drop. User have ability to hear the correct pronunciation of the letter if needed. After correctly pairing the letters user have to pronounce the letter correctly. ARUNALU provides a reward system to encourage the user. Vocal instructions are given by the system in each step.

In the intervention process, there are three levels as letters map, two letter words map, and three letter words map by increasing the difficulty of the levels.

C. Vocal pattern analysis through motion pictures (Reading Environment with set of motion pictures)

In this solution, System will display a basic motion picture after user select the level of identification path (Figure 6). The video is having a basic sentence which is describe the content of the video. User should read that sentence loud. System will provide three circumstances for the dyslexia identification methodology and end of the flow it will display summary of a child. For intervention flow. Application is having three levels words with two letters, words with three letters, and words with four letters. Each level is having three circumstances and by end of each level system will provide a report by considering the average details from all three circumstances. To check whether the word in the sentence is correct or not, user's voice is getting as an input and getting the time that user takes to read the sentence. Those inputs are going through natural language processing. The interaction between human and machine could go as follows,

1. Human talks to the machine
2. The machine captures the audio
3. Audio to text conversion takes place
4. Processing of the text's data
5. Data to audio conversion takes place

Identify the level of dyslexia by comparing the original sound with the user input sound and time takes by user. Then that information is saved in the database for analysis process.

Intervention process also done using an advancement of using set of motion pictures. There, two letter words, three letter words and four letter words will be using to provide better and effective phonological training to the user.



Figure 6: Screening process using set of motion pictures



Figure 7: Intervention process using set of motion pictures

D. Screening and Intervention through voice pattern analysis via Story Building concept (Reading Environment which includes story building and reading)

[18] Reveals that the effectiveness of the storytelling and story building methodologies, to improve the comprehensive capabilities of young children. According to [19], the use of multimedia story applications on touch-interactive mobile devices shows effective results on young children. Therefore, in this component story building methodology has been used as the screening and intervention method. In this solution, the reading and pronunciation tests are carried out as 3-4 sentence storytelling. User can read the given sentences related to the given picture which includes two letter words, three letter words and four letter words. In this special concern is on the letters and words which dyslexic people most struggling and the words which can be read by age 5-6 children.



Figure 8: screening process using story building concept



Figure 9: intervention process using story building concept

For the screening process (Figure 8), user has to read the story word by word given in the reading environment, timer shows the time spent for the reading task. After completing the task, user can see what is the time duration, misspelled words and accuracy level.

For the intervention process (Figure 9), stories which includes two letter words, three letter words, and two- three-four mix letter words applied in the solution. The stories consist of 3-4 sentence content, otherwise it will be too complicated for age 5-6 children.

In these four components Vocal instructions are given to provide a better understanding and help the user to interact with the solution because it takes time to read and understand the instructions.

The model which applied for all the four components consist of data collection of 1800 non-dyslexic school children who are in age 5-6 group. This data collection consist of voice data which includes letters and words, dyslexic children most struggle to read and also considering the letters and words which can be read by children who are in the particular age group. This data collection is done in different regions of Sri Lanka. First the noise removal of the voice clips has to be done due to the background noise. Speech Recognition Speech-to-Text conversion done using Google Speech Recognition API and the Feature extraction was done by Mel Frequency ceptral Coefficient (MFCC). Model training was done using Convolutional Neural Networks (CNN) with 11 layers. Number of epochs used were 100. Several features extracted from input speech signals, through MFCC using for stage Identification of Dyslexia. Inputs of user feeding to the model in the dyslexia screening. Support Vector Machine (SVM) is applied to predict whether the user is Dyslexic or non-dyslexic.

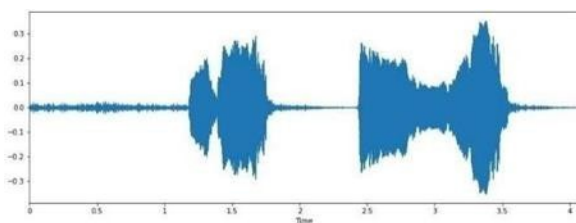


Figure 10: Feature extracted speech signal form

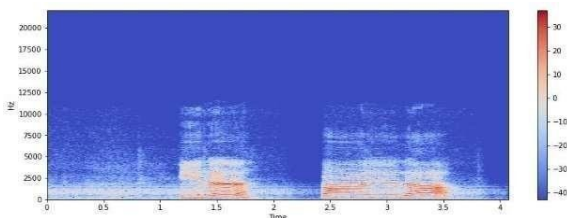


Figure 11: Spectrogram

Four Dyslexia Intervention methodologies are used to provide the phonological support, if the user is screened as Dyslexic or there's a risk of being Dyslexic. User can use preferred intervention methodology and the progress report can be monitored weekly. Therefore user can select the effective intervention method for themselves by analyzing the progress. The Reason for proving four intervention methodologies to provide a unique and effective phonological support and comprehension training for the user.

IV. RESULTS AND CONCLUSION

The main objective of "ARUNALU" is to provide unique and better phonological supportive methodologies, who are having Sinhala reading weaknesses due to Dyslexia. Therefore the solution has been tested among children who are having reading difficulties from different regions of country who are in age 5-6 group. 1500 speech data gathered for letters and words from non dyslexic children in Sri Lanka are fed as training data in the data collection, we've been cautious to apply words which can be read by children in age 5-6 group. Feature extraction has been done using MFCC (Mel-frequency ceptral coefficients) technique and the model training was done using CNN (Convolutional Neural Networks). Obtained 72% of accuracy and it can be improved feeding more training data to the neuralnetwork.

Four screening methods has been tested among 40 students (Age 5-6) which is a mixed sample of Dyslexic and non-Dyslexic children. It obtained 70% of accuracy of categorizing and identifying the risk of Dyslexia, who are pre identified as Dyslexic children from the mixed sample. Using the reading environment and gaming environment for screening and intervention process is a scientifically proven efficient methods has been used in the ARUNALU solution.

During the study, it has been recognized the nature of automatic speech recognition systems (ASR) has been changed into in improvement using Convolutional Neural Network. Application of deep learning approaches can replace the functionality of traditional ASR systems. It can be recognized three main types of architectures for ASR as [20] attention-based methods, connectionist temporal classification and convolutional neural network-based direct raw speech model. In this solution, the approach has been used is convolutional neural network-based direct raw speech model. Obtained 72% of accuracy in CNN model and classification of Dyslexic or Non-Dyslexic has been obtained 85% using Support Vector machine (SVM).

As a result, it seems children were interested in engaging with solution because of the colors, pictures and graphics and the compatible and user friendly environment. And the font and the letter shape (Rounded shape) applied in the solution was similar to Sinhala alphabetic book (*Sinhala Hodi potha*) so it was easier to recognize the letters and words for children who were in age 5-6 group.

The content used in the solution(words, stories) were taken from Teacher's guide for grade one students therefore practicing the words and content through this solution seems effective for their regular study activities. And the guidance through vocal instructions were effective towards the users and received favorable responses.

Therefore, the environment of the application, prove the user friendliness among the children who require the phonological support through this solution. And the phonological training provided in native language (Sinhala) seems very much effective, because most of the children who suffer from Dyslexia in age 4-5 group speak in Sinhala language (Mainly focusing on local context in Sri Lanka). Providing the report analysis through the solution also provides positive impact, because 1. User/ Parent/Teacher can identify and analyze the child's improvement of the training among the time period 2. Parent/ Teacher can identify which intervention methodology is most effective for the child and continue with it and obtain more effective results.

To conclude, "ARUNALU" is the first ever learning ecosystem which provide four screening and intervention methodologies unique from each other, to overcome Sinhala reading weaknesses due to Dyslexia for children in age 5-6 which required early identification. As future task, it's expected to feed more data and obtain a higher accuracy and provide even better version of the solution. Since this solution only developed for Android operating system, it's expected to develop a version for iOS as a future improvement. Since this solution focusing on overcoming reading weaknesses due to Dyslexia, it's expected to provide unique screening and intervention methodologies to overcome Sinhala writing weaknesses due to Dyslexia. This solution focusing on Sinhala language, but we identified there are children who speaks Tamil and English as their native language. Therefore as a future task, it's expected to expand this solution in Tamil and English languages. The ultimate goal of this solution is to provide this for the needy (Dyslexic community) and help the early identification to eradicate Dyslexia from Sri Lanka and support the future generation to achieve their dreams and goals without obstacles.

ACKNOWLEDGMENT

Authors would like to thank Sri Lanka Institute of Information Technology for the encouragement and guidance to pursue this research. And we are thankful for Dr. Kamalini Wanigasighe at Senehasa Research and Information Center (SERRIC) and the staff, for the enormous support while the study. And extreme gratitude towards Capt. W.B.G.Kumuduni, Madam Principle of Rathnawali Balika Vidyalaya - Colombo, Mrs. Anuruddhika Kandakkulama, Madam Principle of Sri Rahula Primary School - Eppawala and school children who participated for data gathering for the research. Furthermore, authors are thankful for the support, comments and advices of academic professionals from Faculty of Computing, Faculty of Business and colleagues at Sri Lanka Institute of Information Technology. The support received from all other parties is acknowledged as well.

REFERENCES

- [1] Singleton C. H., Dyslexia: Cognitive factors and implications for literacy. In G. Reid and J.Wearmouth (Eds.) *Dyslexia and Literacy: Research and Practice*. London: Wiley, pp. 115-130. 2002.
- [2] Weekly Epidemiological Report by Ministry of Health, Vol 42 No 07. 2015.
- [3] SLD (Specific Learning Disabilities) Leadership Committee in conjunction with the Division of Special Education, *Dyslexia Information Paper*. [Accessed March, 2020]
- [4] Barbara Penolazzi, Claudio Vio3 and Alessandro Angrilli, Cortical reorganization in dyslexic children after phonological training: evidence from early evoked potentials Chiara Spironelli. [Accessed March, 2020]
- [5] Schneider W., Roth E., Ennemoser M., Training phonological skills and letter knowledge in children at risk for dyslexia: A comparison of three kindergarten intervention programs. *Journal of Educational Psychology*, 92(2), 284-295. 2000.
- [6] Sally E. Shaywitz, A new model of this reading disorder emphasizes defects in the language-processing rather than the visual system. It explains why some very smart people have trouble learning to read. [Accessed March, 2020]
- [7] Guylaine Le Jan, Régine Le BouquinJeannès , Nathalie Costet , Gérard Faucon, "Discriminatory validity of dyslexia screening tasksin French school age children," August, 2007.
- [8] Paraskevi Tzouveli, Andreas Schmidt, Michael Schneider,AntonisSymvonis, Stefanos Kollias, Adaptive Reading Assistance for the Inclusion of Students with Dyslexia:The AGENT-DYSL approach. July, 2008.
- [9] Kularathna, T. C. P., Kuruppuarachchi, L. C., Wijekoon, W. M. S. S., Kulamini, W. A. M., Samarasingha, R. N., & Tharanga, D., Yalu: Computer Game Based Solution to Screen Learning Disabilities in Kids. 2014.
- [10] S. Thelijjagoda, M. Chandrasiri, D. Hewathudalla, P. Ranasinghe, I.Wickramanayake, "The Hope: An Interactive Mobile Solution to Overcome the Writing, Reading and Speaking Weaknesses of Dyslexia," 2019.
- [11] Barun Paudel , K.M.D.M Karunarathna, "D-Care: A mobile- based therapy solution for dyslexic children between age-group 5-12", 2015.
- [12] L. Rello, C. Bayarri, and A. Gorriaz, "What is wrong with this word? dysegxia," in *Proceedings of the 14th international ACM SIGACCESS conference on Computers and accessibility -ASSETS '12*, 2012, pp. 219-220.
- [13] R. Skiada, E. Soroniati, A. Gardeli and D. Zissis, "EasyLexia: A Mobile Application for Children with Learning Difficulties", 2014.
- [14] L. Marcos, A. Dominguez, J. S. Navarrete and C. Pages, "An empirical study comparing gamification and social networking on e-learning", 2014, pp.82 - 91
- [15] D. Gooch, A. Vasalou, and L. Benton, "Exploring the use of a Gamification Platform to Support Students with Dyslexia", 2015.
- [16] P. Ratnayake, R. Kariyawasam, M. Nadeeshani, T. Hamid and I. Subasinghe, "A Gamified Approach for Screening and Intervention of Dyslexia, Dysgraphia and Dyscalculia", 2019.
- [17] B. W. Betts, J. Bal, and A. W. Betts, "Gamification as a tool for increasing the depth of student understanding using a collaborative e- learning environment," *International Journal of Continuing Engineering Education and Life-Long Learning*, 2013.
- [18] I. Rebecca, S. Joseph, L. Liane and L. April, "The Effects of Storytelling and Story Reading on the Oral Language Complexity and Story Reading on the Oral Language Complexity and Story Comprehension of Young Children" *Journal of Early Childhood Education*, Dec., pp.157-163, 2004.
- [19] Z. Ninger and Y. Aman, "Effects of Multimedia Story Reading and Questioning on Preschoolers' Vocabulary Learning, Story Comprehension and Reading Engagement" *Journal of Educational Technology Research and Development*, July, 2017.